

Capstone Project -17

modular, interconnected capstone project framework for *Smart City Traffic & Pollution Analytics*. Each section has **10 complex, scenario-based questions** arranged in the order you requested:

Scenario: Kolkata Municipal Corporation deploys IoT sensors across 50 major junctions to monitor **traffic congestion** and **pollution levels (PM2.5, NO2, CO)**. During festive seasons (e.g., Durga Puja), traffic density spikes by 60–70%, while AQI levels rise sharply. The city wants to:

- Validate sensor data accuracy (Excel)
 - Aggregate and query IoT data (SQL)
 - Perform time-series analysis (Python)
 - Apply statistical analytics without ML (Data Science)
 - Build a real-time monitoring dashboard (Power BI)
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1. Advanced Excel (Trend Validation & Pre-Analysis)

1. Using pivot tables, how can you validate hourly traffic volume trends across multiple junctions?
 2. Create a dynamic chart comparing pollution levels (PM2.5, NO2) with traffic density over weekdays vs weekends.
 3. How would you use conditional formatting to highlight traffic congestion spikes exceeding 80% capacity?
 4. Design an Excel dashboard that validates IoT sensor data consistency across 5 different city zones.
 5. Apply advanced formulas (INDEX-MATCH, OFFSET) to cross-check pollution readings against traffic flow.
 6. How can you use Excel's Data Validation to restrict erroneous sensor entries (e.g., negative traffic counts)?
 7. Build a scenario analysis model to compare traffic congestion during peak vs off-peak hours.
 8. Use Excel Solver to optimize traffic flow distribution across three major highways.
 9. How would you validate time-series gaps in pollution data using Excel functions?
 10. Create a consolidated Excel report that flags anomalies in both traffic and pollution datasets.
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2. SQL (IoT Sensor Data Management)

1. Write a query to aggregate hourly traffic counts from IoT sensors across all junctions.
2. How would you join traffic and pollution tables to analyze correlation by timestamp?
3. Create a query to detect missing sensor readings in traffic data.

4. Write a SQL query to rank top 10 most congested junctions in the city.
 5. How would you use window functions to calculate moving averages of pollution levels?
 6. Query to compare traffic density between weekdays and weekends.
 7. Write a query to detect outliers in pollution readings (e.g., > 500 AQI).
 8. How can you partition traffic data by zone and calculate congestion ratios?
 9. Create a query to validate sensor uptime percentage across all IoT devices.
 10. Write a query to generate a combined dataset for Power BI ingestion (traffic + pollution).
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3. Python (Time-Series Analysis)

1. How would you use Pandas to resample traffic data into 15-minute intervals?
 2. Write Python code to detect missing timestamps in pollution data.
 3. Use Matplotlib to plot traffic vs pollution trends over a week.
 4. How can you apply rolling averages to smooth traffic congestion data?
 5. Write a script to calculate correlation coefficients between traffic and pollution.
 6. How would you handle time-zone inconsistencies in IoT sensor data?
 7. Use Python to generate a comparative heatmap of traffic density across city zones.
 8. Write code to detect anomalies in pollution readings using statistical thresholds.
 9. How can you merge traffic and pollution datasets for synchronized analysis?
 10. Use Python to export validated datasets into Excel for cross-checking.
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4. Data Science (Without ML, Pure Analytics)

1. How would you design a framework to measure the impact of traffic on pollution levels?
2. Apply statistical hypothesis testing to validate whether traffic congestion significantly increases pollution.
3. Use regression-free correlation analysis to compare traffic density vs AQI.
4. How can you apply clustering logic (without ML) to group junctions by traffic intensity?
5. Design a KPI framework for Smart City monitoring (traffic flow, AQI, congestion ratio).
6. How would you calculate seasonality effects in traffic data (e.g., festive vs normal days)?
7. Apply variance analysis to detect fluctuations in pollution levels across zones.
8. How can you design a congestion index combining traffic and pollution metrics?
9. Use time-series decomposition (trend, seasonality, residual) for traffic data.

10. Create a structured data pipeline integrating SQL, Python, and Excel outputs.
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5. Power BI (Real-Time Monitoring Dashboard)

Dashboard Theme: Smart City Traffic & Pollution Analytics

- **Main KPIs:** Traffic congestion %, AQI levels, sensor uptime, zone-wise comparison.
- **Visuals:**
 - Line chart: Traffic vs Pollution over time
 - Heatmap: Zone-wise congestion intensity
 - Gauge: AQI levels compared to safe thresholds
 - Card visuals: Top 5 congested junctions
 - Real-time streaming tiles: Sensor uptime %

10 Power BI Questions:

1. How would you design a real-time dashboard integrating SQL and Python outputs?
 2. Create a Power BI measure to calculate congestion ratio dynamically.
 3. How can you use DAX to calculate rolling averages of AQI?
 4. Design a drill-through report to analyze traffic at junction-level.
 5. How would you implement real-time streaming datasets in Power BI?
 6. Create a KPI card showing % increase in pollution during peak traffic hours.
 7. How can you use bookmarks to toggle between traffic and pollution views?
 8. Design a Power BI dashboard to highlight anomalies detected in Python.
 9. How would you integrate Excel validation reports into Power BI visuals?
 10. Create a unified dashboard combining traffic, pollution, and sensor uptime metrics.
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This structure ensures **interconnected flow**:

- Excel validates → SQL aggregates → Python analyzes → Data Science interprets → Power BI visualizes.